

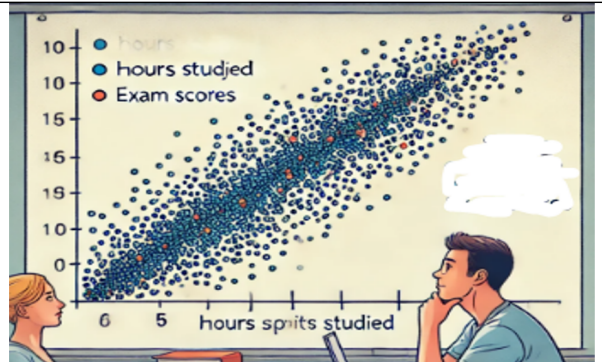
UNIT 2 MCQs

Scatter plots, correlation coefficient and its properties

1.

Q: A researcher collects data on the number of hours studied and exam scores of 50 students. The scatter plot of the points shown in the figure. Which of the following best describes the correlation between hours studied and exam scores?

- A) Strong negative correlation
- B) Strong positive correlation
- C) No correlation
- D) Weak negative correlation



2.

Q 1.16. If we draw a graph with the help of following data in the given table, then what type of correlation between X and Y :

X	10	20	30
Y	70	140	210

- (a) Perfect positive correlation.
- (b) Perfect negative correlation.
- (c) No correlation.
- (d) Can not say any thing from the given data.

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5. If all points on a scatter diagram lie on a perfectly straight line falling from the top left to the bottom right, the correlation is:

- a) Perfect positive
- b) Perfect negative
- c) Zero
- d) Indeterminate

4.

Q1. The correlation coefficient is unaffected by:

- A) Change in origin
- B) Change in scale
- C) Both origin and scale
- D) Change in units only

5.

Q2. In a scatter plot, each point represents:

- A) The mean of the data set
- B) A pair of values (x, y) for two variables
- C) A frequency of one variable
- D) A standard deviation

6.

Q1: **If X and Y are independent random variables, then the correlation between them is:**

- A. 1
- B. -1
- C. 0
- D. Undefined

Karl Pearson's correlation coefficient

7.

For two correlated variables x and y, if coefficient of correlation between x and y is 0.8014, variance of x and y are 16 and 25 respectively. Then the covariance between x and y is:

1. 162.08

2. 16.028

3. 160.28

4. 16.208

8.

Limits for correlation coefficient.

(a) $-1 \leq r \leq 1$

(b) $0 \leq r \leq 1$

(c) $-1 \leq r \leq 0$

(d) $-1 \leq r \leq 2$

9.

3. The formula for Karl Pearson's correlation coefficient is:

A) $r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

B) $r = \frac{\frac{\sigma_X \sigma_Y}{\text{Cov}(X, Y)}}{\frac{X - \bar{X}}{Y - \bar{Y}}}$

C) $r = \frac{X - \bar{X}}{Y - \bar{Y}}$

D) $r = \text{Cov}(X, Y) \times \sigma_X \sigma_Y$

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4: If $X = [2, 4, 6, 8]$ and $Y = [5, 10, 15, 20]$, find the correlation coefficient.

A. 1

B. 0.9

C. 0

D. 0.5

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The correlation coefficient $r(X, Y)$ is 0.6. Find $r(U, V)$, where $U = \frac{X-2}{5}$ and $V = \frac{Y-1}{7}$.

(a) 0.4 (b) 0.66 (c) 0.6 (d) 0.5

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The covariance between X and Y is 0.35, variance of X is 1.1576 and variance of Y is 1.6075. Find the correlation coefficient.

(a) 0.25 (b) 0.35 (c) 0.4 (d) 0.5

Spearman's rank correlation coefficient

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If rank correlation between two data sets is $\frac{3}{7}$ and $\sum_{i=1}^n d_i^2 = 20$ then calculate the sample size n .

(a) 10 (b) 5 (c) 6 (d) 8

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If $\sum(X - \bar{X})(Y - \bar{Y}) = 150$, $\sum(X - \bar{X})^2 = 100$, and $\sum(Y - \bar{Y})^2 = 225$, then what is the value of Karl Pearson's coefficient of correlation (r)?

- a) 0.5
- b) 0.9
- c) 1.0
- d) 0.7

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If $d^2 = 0$, then rank correlation is

- (a) 0 (b) 1 (c) 0.5 (d) -1

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3. The formula for Spearman's rank correlation coefficient (when no tied ranks exist) is:

- A) $\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$
- B) $\rho = \frac{\sum d^2}{n(n^2 - 1)}$
- C) $\rho = \frac{6 \sum d^2}{n(n^2 - 1)}$
- D) $\rho = 1 + \frac{6 \sum d^2}{n(n^2 - 1)}$

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4: Given the ranks:

X 1 2 3 4 5

Y 2 1 4 3 5

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Find rank correlation

- A. 1.0
- B. 0.9
- C. 0.8
- D. 0.7

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For Spearman's rank correlation, if the correlation coefficient is 0.7 and $\sum_{i=1}^n d_i^2 = 49.5$ then the value of sample size 'n' is:

- a) 99 b) 20 c) 10 d) 990.

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The Rank of 4 students in Maths and Physics are as follows. Two members with brackets denotes the rank of students in Maths and Physics (1,1), (2,4), (3,2), (4,3). Then the rank correlation coefficient is

- (a) 0.1 (b) 0.2 (c) 0.4 (d) 0.8

Linear regression and its properties

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Given the two lines of regression as $3X - 4Y + 8 = 0$ and $4X - 3Y = 1$, the means of X and Y are

- a) $\bar{X} = 5, \bar{Y} = 2$, b) $\bar{X} = 5, \bar{Y} = 4$, c) $\bar{X} = 4, \bar{Y} = 5$, d) $\bar{X} = 7, \bar{Y} = 3$

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If the regression line is $Y = 3 + 2X$, what is the predicted value of Y when $X = 5$

- (a) 10
(b) 12
(c) 13
(d) 14

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If two regression coefficients are $b_1 = 4/5$ and $b_2 = 9/20$. Then the value of correlation coefficient r is

- (a) 2 (b) 0.6 (c) 1.5 (d) -1

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If the correlation coefficient between X and Y is 0.8, then the correlation coefficient between $U = (X-10)/2$ and $V = (Y-5)/2$ is

- (a) 0.8 (b) 0.016 (c) 0.4 (d) 0.2

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If X and Y are random variables with variances 36 and 16, respectively, then the correlation between $U=X+Y$ and $V=X-Y$ is

- (a) 1 (b) $5/13$ (c) $13/20$ (d) 0

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If X and Y are correlated variables then $E(Y/X)$ and $E(X/Y)$ will represent

- (a) **The Regression curves**
(b) The Correlations
(c) The Normal equations
(d) None of the above

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The regression equations of two variables X and Y are $X = -0.7Y + 5.2$ and $Y = -0.3X + 2.8$. Then the coefficient of Correlation between X and Y is

- (a) -0.7
(b) 0.46
(c) -0.21
(d) -0.46

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If $X = 4Y + 5$ and $Y = kX + 4$ are the regression lines of X on Y and of Y on X then

- (a) $0 \leq k \leq 1$
(b) $0 \leq k \leq 1/2$
(c) $0 \leq k \leq 1/4$
(d) $-1 \leq k \leq 1$

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Given the correlation coefficient between X and Y is 0.533 and the standard deviation of X is 5. If one of the regression coefficients is 0.2, then the standard deviation of Y is

- (a) 1.42
(b) 15.32
(c) 13.32
(d) None

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If the correlation between X and Y is 0.4, then the correlation between $2X$ and $2Y$ is

- (a) 0.2
(b) 0.16
(c) 0.4
(d) 0.8

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☐ In the regression line $Y = a + bX$, where $\bar{X} = 2.5$, $\bar{Y} = 5.5$ and $a = 1.50$, then value of b is

- (A) 1.75 (B) 1.60 (C) 2 (D) 2.5

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☐ Two regression lines are given as $3X + 2Y = 26$, $6X + Y = 31$, then ratio of variances of X to Y is

- (A) $\frac{1}{9}$ (B) $\frac{1}{4}$ (C) $\frac{2}{5}$ (D) $\frac{3}{8}$

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☐ Angle between the regression lines for two uncorrelated variables is

- (A) 0° (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

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☐ If the regression coefficients of regression equation of X on Y is 0.4 and of Y on X is 1.6, then the regression coefficient of $U=3X$ on $V=2Y$ is

- a) 0.4 b) 1.6 c) 1.066 d) 0.6

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7. The regression coefficient of Y on X is given by:

- A) $b_{yx} = \frac{r\sigma_Y}{\sigma_X}$
 B) $b_{yx} = \frac{r\sigma_X}{\sigma_Y}$
 C) $b_{yx} = \frac{\sigma_Y}{\sigma_X}$
 D) $b_{yx} = r^2$

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9. The product of the two regression coefficients equals:

- A) r
 B) r^2
 C) r^3
 D) $r/2$

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If $b_{yx}=1.5$ and $r=0.75$, then $b_{xy}=?$

- A. 0.5
- B. 1.0
- C. 0.375
- D. 0.25

Q 38

Q.

If $y = -10x$ and $x = -0.1y$ then r is equal to –
(a) 1 (b) -1 (c) 0.1 (d) None of these-

Q.39

If unexplained variation=19.22 ,explained variation=19.70 then coefficient of determination is -

- (a) 0.5062 (b) ± 0.3123
- (b) ± 0.1234 (d) 0

Q.40

If the regression coefficient of X on Y is $-1/6$ and that of Y on X is $-3/2$ then coefficient of correlation is

- (a) - 0.5 (b) 1 (c) -1 (d) None of these